BIRMINGHAM CITY
University

CMP6200
Individual Undergraduate Project
Project Interim Report
BCG: A Game-Based Approach to Motor Imagery Adaptation and BCI Calibration

Student Name: Cao Đăng Khoa

Student ID: 25195675

Course: Bachelor of Science with Honours Computer Science

Supervisor: Dr. Vi Chí Thành

Date: December 8, 2025

Table of Contents

List of Figures

List of Tables

1.0 INTRODUCTION AND CONTEXT

1.1 Background and Significance

The way humans interact with computers has undergone a significant evolution over time. From early keyboards and mice to modern touchscreen devices and voice assistants, each innovation has expanded how people can engage with digital systems. But what if humans were able to communicate with technology simply by thought? This is what next-generation technology is achieving: Brain-Computer Interfaces (BCIs), systems that allow people to control devices with their thoughts (Birbaumer, 2006). By reading neural signals directly from the human brain and translating them into digital commands for computers to understand, BCIs not only enhance the experience for typical users but also enable people with physical limitations to access technology more easily.

Among various types of BCI applications, electroencephalography-based motor imagery (EEG-MI) is one of the most widely studied areas (Lotte et al., 2018). This concept is based on a key principle of neuroscience: whenever a person imagines a movement, such as moving a hand or foot, without performing it physically, their brain still produces characteristic signals with patterns that are remarkably similar to the actual physical execution (Neuper and Pfurtscheller, 2001). Building on this principle, EEG-based MI has been applied in many real-world scenarios, including motor rehabilitation, assistive control for individuals with disabilities, and control of games (Pfurtscheller and Lopes da Silva, 2001).

These promising applications, however, still depend heavily on how accurately MI signals can be detected. This accuracy is influenced by many factors, with the calibration process playing a crucial role, which is also the main focus of this project (Lotte et al., 2018).

1.2 Current State and Limitations

In controlled laboratory environments, where the calibration process is managed with precision, EEG-MI systems demonstrate a high level of performance in classifying motor signals (Lotte et al., 2018). After an initial training phase, these classifiers often reach accuracies above 70% in distinguishing between imagined movements such as left hand, right hand, foot, or a rest state, a level that is commonly taken as the threshold for effective BCI control (Kübler et al., 2001).

However, achieving this level of performance for each individual user requires a significant calibration effort. In practice, the system must learn a user's specific neural patterns through 15-30 minutes of supervised training, typically collecting 40-80 labelled trials per movement class, around 120-240 trials in total (Giles et al., 2022). Moreover, these EEG patterns vary significantly between sessions even for the same user due to non-stationarities (Sugiyama et al., 2008).

The repetitive and cognitively demanding nature of EEG-MI makes it challenging to implement outside controlled laboratory settings (Xu et al., 2024). It is time-consuming, requires dedicated human supervision, and depends on specialised lab facilities - which are often geographically distant. It also induces cognitive fatigue, degrades signal quality over time, and limits practicality for multi-session applications like rehabilitation (Xu et al., 2024). Consequently, despite decades of research, this technology remains largely confined to laboratories rather than achieving widespread real-world adoption (Wolpaw et al., 2002).

1.3 Problem Statement

EEG-MI calibration per session requires 15-30 minutes of repetitive trials, which limits real-world scalability despite over 70% success in the experimental environment (Giles et al., 2022; Kueper et al., 2024). This limitation arises from changes in brain signals between sessions and across different individuals (Sugiyama et al., 2008), the need for human supervision (Giles et al., 2022), and reliance on specific infrastructure (Lotte et al., 2018).

2.0 REVIEW OF EXISTING KNOWLEDGE

2.1 The Neuroscience of Motor Imagery

To interpret human intention, we must examine the concept of functional equivalence. Modern research has shown that planning a movement and physically performing it are not distinct processes (Jeannerod, 1995). When a person performs Motor Imagery, mentally imagining a movement while remaining physically still, their brain activates many of the same neural networks used during actual execution.

When a user imagines moving a limb, such as their right hand, the primary motor cortex generates specific, rhythmic electrical patterns. These patterns occur in the sensorimotor cortex and are governed by the contralateral principle: imagining a right-hand movement primarily activates the left hemisphere, while left-hand imagery activates the right (Wolpaw et al., 2002).

These activations are measurable through changes in brain rhythms. In a resting state, neurons fire in synchrony, creating strong idle rhythms known as Mu (8-13 Hz) and Beta (13-30 Hz). When the user begins imagining movement, this synchrony is disrupted, causing a power drop known as Event-Related Desynchronisation (ERD). Conversely, when the brain relaxes, the neurons synchronise again, leading to Event-Related Synchronisation (ERS) (Pfurtscheller and Lopes da Silva, 2001).

These ERD/ERS patterns are the core signatures BCI systems are trained to recognise. While they are highly reproducible and occur even in individuals with severe motor disabilities, their specific structure varies significantly between individuals (Lotte et al., 2018). The frequency and spatial distribution of these signals are unique to each person, necessitating a system that can adapt to the user.

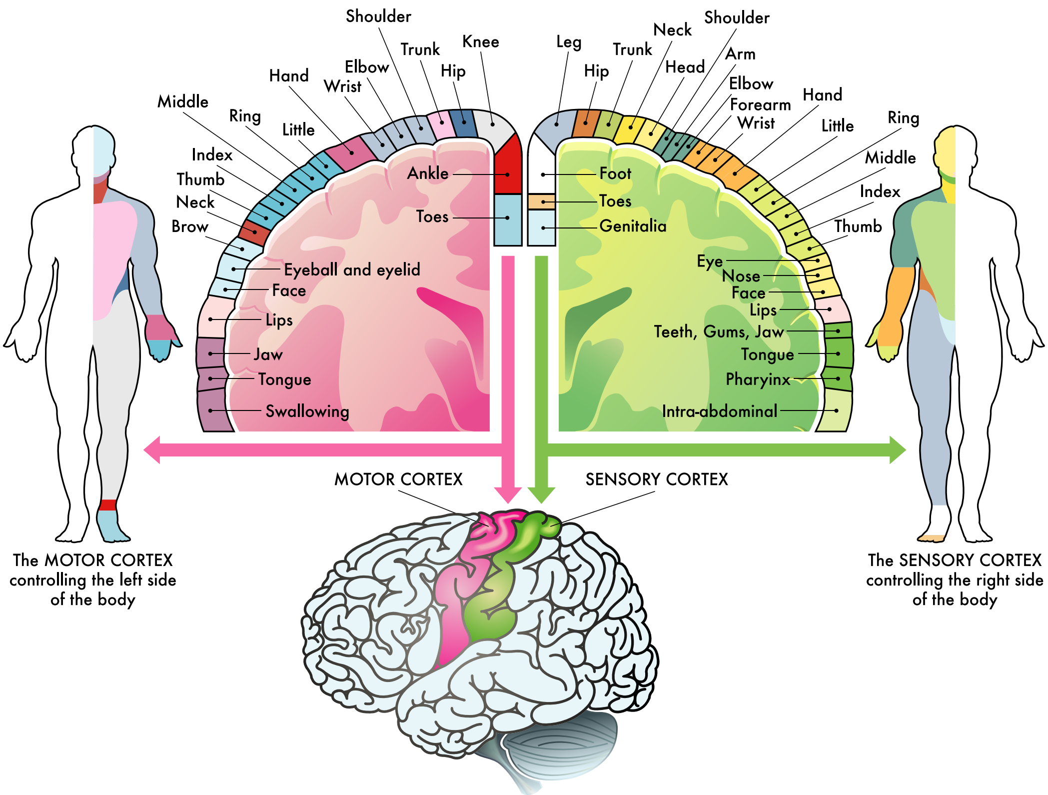

Figure 2.0.1: Motor Cortex Organization and Contralateral Hemispheric Control

2.2 The EEG Processing Pipeline

Capturing these neural signatures non-invasively is a significant engineering challenge. Raw scalp EEG is weak, often in the microvolt range, and obscured by noise such as muscle movement and electrical interference (Niedermeyer and Lopes da Silva, 2004). Therefore, data must pass through a strict processing pipeline.

The first stage is preprocessing. Since motor imagery activity is concentrated in the Mu and Beta bands (8-30 Hz), bandpass filters are applied to isolate these frequencies, removing high-frequency muscle noise and slow baseline drifts. Following this, spatial filtering is used to boost the signal-to-noise ratio. This mathematically separates mixed signals from multiple electrodes, isolating activity from the sensorimotor cortex while suppressing noise from unrelated brain regions.

The next stage is feature extraction. The well-known algorithm for this is Common Spatial Patterns (CSP) (Blankertz et al., 2008). CSP functions like a sophisticated filter that combines signals from different electrodes to create new channels. Its goal is to find a specific blend that maximises the variance of signal power for one task (e.g., right hand) while minimising it for another. This transforms complex biological signals into separable features, simplifying the prediction task for the computer.

2.3 Classification Strategies

Once features are extracted, they are fed into a classifier to determine the user's intent. For years, Linear Discriminant Analysis (LDA) and Support Vector Machines (SVM) have been the standard classifiers (Lotte et al., 2018). They remain popular due to their computational efficiency on small datasets, making them ideal for real-time applications.

Recently, the field has explored Deep Learning techniques, such as Convolutional Neural Networks (CNNs) (Raza et al., 2019). These models promise to automatically learn features from raw data, potentially bypassing manual steps like CSP. However, Deep Learning models typically require thousands of trials to perform reliably (Roy et al., 2019). This is often impractical for real-time BCI usage, where setup time must be minimised. Therefore, traditional pipelines using CSP and LDA remain the most practical choice for achieving high accuracy with limited calibration time.

2.4 The Calibration Bottleneck

Despite advanced algorithms, a significant obstacle prevents widespread adoption: the dependency on long and repetitive calibration. Current high-performing BCI systems typically use the "Standard Graz Protocol", where the user follows a fixed sequence of abstract prompts (e.g., arrows on a screen) for 20 to 30 minutes (Pfurtscheller and Lopes da Silva, 2001).

This phase is compulsory because brain signals are non-stationary. They change constantly due to fatigue, attention levels, and electrode shifts (Sugiyama et al., 2008). A classifier trained on Monday could fail on Tuesday because the statistical distribution of the data has shifted. Consequently, the system must essentially re-learn the user at the start of every session.

This process creates a severe bottleneck. The abstract nature of the protocol induces boredom and cognitive fatigue (Xu et al., 2024). As the user tires, their ability to generate distinct neural patterns degrades, leading to poor quality data. This creates a cycle where poor data leads to a poor classifier, frustrating the user and further degrading signal quality.

2.5 Gamification in BCI

Gamification applies game-design elements to non-game contexts to enhance motivation. In BCI, this is typically utilised during Neurofeedback training (Škola and Tinková, 2019).

Instead of abstract cues, users interact with compelling elements, such as controlling a race car. Research confirms that this approach improves signal quality; when a user is motivated and enters a flow state, their attention focuses, producing more pronounced ERD/ERS patterns (Atilla, 2024).

However, the application of gamification remains uneven. While the training phase has evolved, the critical calibration phase often retains rigid, repetitive protocols. Users must endure a fatiguing setup before accessing the engaging interface, leading to a loss of engagement exactly when the system needs high-quality data.

This presents a significant opportunity: by applying engagement mechanics directly to the calibration phase, it is theoretically possible to collect high-quality data. This would transform calibration from a tedious setup barrier into a seamless part of the user experience. However, the game design process is critical. If a game is visually chaotic, it may generate interfering signals. The design must balance engagement with signal purity.

2.6 Critical Analysis and Project Justification

This review reveals a contradiction in BCI research: theoretical foundations are strong, but practical application is limited by the calibration bottleneck. Mathematical solutions like Transfer Learning address the lack of data but struggle with biological non-stationarity (Sugiyama et al., 2008). Conversely, gamification improves signal quality but is rarely applied to the calibration phase itself.

Therefore, a significant gap exists: there is a lack of frameworks that replace the standard, repetitive calibration protocol with an engaging, game-based alternative.

This project directly addresses this gap. By developing a system that uses a game as the data collection task, this project aims to investigate whether a Gamified Calibration Protocol (playing to calibrate) can effectively replace the traditional Standard Graz Protocol. If successful, it will demonstrate that high-quality calibration data can be collected more efficiently while significantly reducing user boredom and cognitive fatigue.

3.0 PROJECT AIMS, OBJECTIVES, AND SCOPE

3.1 Project Aim

The primary aim of this project is to investigate and validate the effectiveness of a gamified calibration protocol compared to the standard clinical approach. Specifically, the project seeks to determine if a game-based interface can reduce user cognitive fatigue and perceived setup time while maintaining sufficient signal quality for BCI control.

3.2 Project Objectives

To achieve this aim, the project focuses on the following measurable objectives, prioritised by their contribution to the comparative study:

1. **Framework Development:** Design and implement a dual-mode BCI software framework that supports two distinct data collection protocols:
    - **Condition A (Standard):** A baseline mode replicating the traditional Graz protocol.
    - **Condition B (Gamified):** A proposed mode where calibration data is collected implicitly during gameplay.
2. **User Experience Evaluation:** Conduct a comparative user study to quantitatively assess the psychological impact of the two protocols. The objective is to measure differences in subjective workload (via NASA-TLX) and boredom levels to determine if the gamified approach effectively mitigates calibration fatigue.
3. **Signal Quality Assessment:** Record and analyse EEG data from both conditions to ensure the gamified approach does not compromise technical performance. The objective is to verify that motor imagery signals (ERD/ERS patterns) generated during gameplay are distinguishable and comparable in quality to those generated during standard training.
4. **Adaptive Implementation:** Develop and integrate an automatic classifier update mechanism within the game loop. This objective focuses on the technical feasibility of retraining the machine learning model (CSP+LDA) in the background using the labeled game data.

3.3 Project Scope

The project boundaries are defined to ensure the feasibility of the comparative study within the academic timeline.

In-Scope:

- Development of the Python-based signal processing backend (preprocessing, feature extraction).
- Design of a 2D/3D game environment (e.g., GameMaker or PyGame) specifically for Motor Imagery tasks.
- Integration with commercially available dry-electrode headsets (e.g., BrainBit).
- Statistical analysis of user fatigue metrics and classification accuracy.

Out-of-Scope:

- Hardware engineering (no sensor modification or headset construction).

- Advanced Deep Learning models (focus remains on CSP+LDA for real-time feasibility).
- Clinical trials with patients (focus is on healthy user validation).
- Long-term adaptation studies (focus is on single-session immediate effects).

4.0 PRODUCT DESIGN

4.1 Design and Methods

The project follows an experimental systems development approach, structured to facilitate the direct comparison of the two calibration methods.

4.1.1 Phase 1: System Development

The software architecture follows a modular design strategy, prioritizing the development of the core comparative framework. This ensures that the primary objective - validating the user experience differences - is achieved early, with advanced adaptive algorithms integrated as a subsequent layer.

The system is built in three main parts. The first two parts form the foundation, while the third part introduces the advanced intelligence needed to automate the calibration.

First, we build the Signal Processing Unit. This Python software connects to the EEG headset and captures the raw brain data. Since raw signals are noisy, this unit applies digital filters (8–30Hz) to strip away muscle noise and electrical interference, ensuring the system receives clean motor imagery signals regardless of which calibration mode is active.

Second, we create the Game Interface. This will be a simple continuous game, such as an infinite runner. The game acts as the data collector. When an obstacle appears (e.g., a rock on the left), it makes the user to imagine a movement. This game event creates a natural “time-stamp” that tells the system exactly when the user was thinking about moving.

Third, we implement the Adaptive Learning Layer, which is the advanced component of the system. Unlike basic BCI games that just read signals, this smart module runs in the background to analyse the labeled data segments from the game. It uses this data to mathematically retrain the computer instantly, allowing the system to adapt to the user’s unique brain patterns automatically without stopping the gameplay.

4.1.2 Phase 2: User Evaluation

The validation phase is the core of the project. We will invite healthy participants ($N = 5 - 10$) to test the system in a Within-Subjects experiment.

Each participant will perform two sessions:

1. **Session A (Traditional-shortened):** The user performs 5 minutes of standard training (watching arrows).
2. **Session B (Standard):** The user performs 15 minutes of standard training (watching arrows).
3. **Session C (Gamified):** The user plays the game for 5 minutes.

By comparing the setup time, accuracy, and reported fatigue between these three sessions, we can validate the project's aim even if the adaptive algorithm requires further refinement.

4.2 Justification of Methods

The comparative design is chosen to directly measure the "Human Factor" benefits of the system. While technical accuracy is important, the primary argument of this thesis is that gamification reduces the barrier to entry. Therefore, measuring Subjective Workload (NASA-TLX) is just as critical as measuring classifier accuracy.

Technically, Common Spatial Patterns (CSP) and Linear Discriminant Analysis (LDA) are selected because they are well-documented and computationally light. This minimises the risk of technical failure during the implementation phase, ensuring the requirements are suitable for an individual project.

4.3 Project Timeline

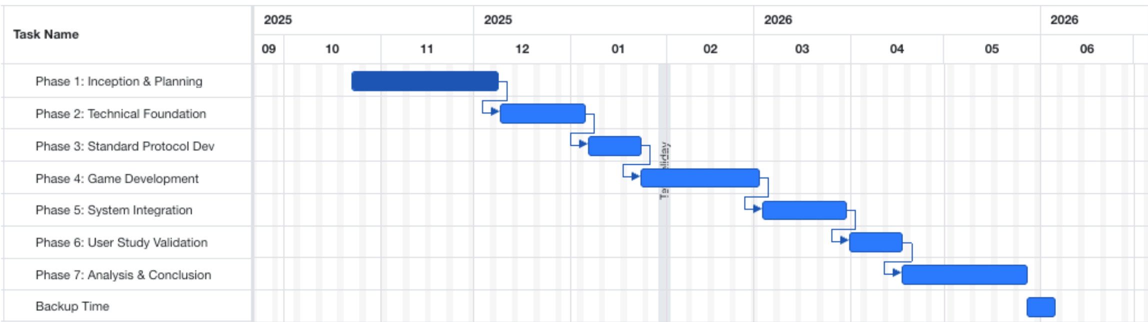

Figure 4.0.1: Project Timeline

5.0 FEASIBILITY, RISKS, AND ETHICAL CONSIDERATIONS

5.1 Feasibility and Resources

5.1.1 Project Feasibility

This project is assessed as highly feasible. By structuring the work around a comparative framework first, the project ensures a valid research outcome (the comparison of user experience) even if the complexity of real-time adaptation presents challenges. The feasibility is further supported by the use of mature Python libraries and the modular design of the software.

5.1.2 Required Resources

The project requires the following resources to ensure successful execution:

- **Hardware:**
  - A 4-channel dry-electrode EEG headband (e.g., BrainBit) with Bluetooth Low Energy (BLE).
  - A standard development laptop with sufficient processing power for real-time streaming.
- **Software:**
  - **Development:** Python 3.9+ and libraries (e.g., MNE-Python, Scikit-learn).
  - **Game Engine:** GameMaker or PyGame.
  - **Version Control:** GitHub will be used for source code management and backup.
  - **Analysis:** Statistical software (e.g., SPSS or Excel) for evaluating user study results.
- **Infrastructure & Environment:**
  - **Testing Environment:** A quiet, isolated room to minimize external signal artefacts and distraction during data collection.
  - **Storage:** Access to University One Drive and secure servers for ethical data management.

5.2 Risks and Mitigation

A comprehensive risk assessment has been conducted to identify potential bottlenecks. The table below outlines these risks and the associated strategies to mitigate them.

Table 5.0.1: Risk Assessment Table

Category	Potential Risk	Impact	Mitigation Strategy
Hardware (Critical)	Electrode placement misses the Central Cortex (C3/C4).	High	Map electrodes early. Use Flex kit or adjust mounting to ensure contact.
Technical	Dry sensors have high noise/impedance.	High	Implement strict artefact removal (8-30Hz) and stillness mechanics.
Project Scope	Adaptive algorithm is too complex for timeframe.	Medium	Focus on the comparative study (Obj 1 & 2) as the core deliverable; treat advanced adaptation as a secondary goal.
Subject	BCI Illiteracy (∼20% of users) (Blankertz et al., 2010).	Medium	Screen participants; exclude non-responders from stats.

5.3 Ethical Considerations

The project adheres to Ethical Application Number 13740. Participants (18+, non-vulnerable) provide informed consent. Data is anonymized, stored on University One Drive, and deleted from local devices within 24 hours. No clinical applications are involved.

References

Atilla, F. (2024). Gamification of motor imagery brain-computer interface training: a systematic review. *Computers in Human Behavior: Artificial Humans*, 2:100041.

Birbaumer, N. (2006). Brain-computer-interface research: coming of age. *Clinical Neurophysiology*, 117(3):479–483.

Blankertz, B., Sannelli, C., Halder, S., Hammer, E. M., Kübler, A., Müller, K. R., Curio, G., and Dickhaus, T. (2010). Neurophysiological predictor of SMR-based BCI performance. *NeuroImage*, 51(4):1303–1309.

Blankertz, B., Tomioka, R., Lemm, S., Kawanabe, M., and Müller, K. R. (2008). Optimizing spatial filters for robust brain-computer interfaces. *IEEE Transactions on Biomedical Engineering*, 55(8):1881–1898.

Giles, J., Ryu, S. H., Kang, P. H., et al. (2022). A transfer learning algorithm to reduce brain-computer interface calibration time. *Frontiers in Neuroergonomics*, 3:837307.

Jeannerod, M. (1995). Mental imagery in the motor context. *Neuropsychologia*, 33(11):1419–1432.

Kueper, N. et al. (2024). Avoidance of specific calibration sessions in motor intention recognition. *Nature Scientific Reports*, 14:28234.

Kübler, A., Kotchoubey, B., Kaiser, J., Wolpaw, J. R., and Birbaumer, N. (2001). Brain-computer communication: unlocking the locked in. *Psychological Bulletin*, 127(3):358–375.

Lotte, F., Bougrain, L., Cichocki, A., Clerc, M., Congedo, M., Kothe, C., and Yger, F. (2018). A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update. *Journal of Neural Engineering*, 15(3):031005.

Neuper, C. and Pfurtscheller, G. (2001). Event-related dynamics of cortical rhythms: frequency-specific features for brain-computer interfaces. *Progress in Brain Research*, 159:199–213.

Niedermeyer, E. and Lopes da Silva, F. (2004). *Electroencephalography: Basic Principles, Clinical Applications, and Related Fields*. Lippincott Williams & Wilkins, 5th edition.

Pfurtscheller, G. and Lopes da Silva, F. H. (2001). Event-related EEG/MEG synchronization and desynchronization: basic principles. *Clinical Neurophysiology*, 110(11):1842–1857.

Raza, H., Rathee, D., Zhou, S. M., Cecotti, H., and Dahal, K. (2019). Deep learning for electroencephalography-based brain-computer interfaces: A systematic review. *Neurocomputing*, 378:108–129.

Roy, Y., Banville, H., Albuquerque, I., Gramfort, A., Falk, T. H., and Fyshe, A. (2019). Deep learning-based electroencephalography analysis: a systematic review. *Journal of Neural Engineering*, 16(5):051001.

Sugiyama, M., Krauledat, M., and Müller, K. R. (2008). Domain adaptation for learning with non-stationary environments. *Neural Networks*, 21(9):1490–1503.

Wolpaw, J. R., Birbaumer, N., McFarland, D. J., Pfurtscheller, G., and Vaughan, T. M. (2002). Brain-computer interfaces for communication and control. *Clinical Neurophysiology*, 113(6):767–791.

Xu, Y. et al. (2024). Improved motor imagery training for subject's self-modulation in EEG-based brain-computer interfaces. *Frontiers in Human Neuroscience*, 18:1447662.

Škola, F. and Tinková, S. (2019). Progressive training for motor imagery brain-computer interfaces using gamification and virtual reality embodiment. *Frontiers in Neuroscience*, 13:1054.